

The Myowearable Sleeve: A Surface Electromyography Injury Prevention Device

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Abstract

Many injured athletes face the possibility of recurring injury while unknowingly overstraining their muscles. To help prevent these, a sleeve-mounted medical device called the Myowearable was developed, able to monitor the amount of electrical stimulation in the adjacent muscle and alert the user when they are in danger of injury. Three electrodes of a surface electromyography (sEMG) sensor touching the skin record the stimulation in the muscle and send the collected data to a paired computer. The software on the computer provides real-time visual and audio feedback to the user, as well as analysis on the collected data to inform them when they are more in danger of hurting themselves.

The current Myowearable prototype was designed to analyze stimulation in one of two easily-accessible muscles depending on where the sleeve is worn: the semimembranosus hamstring tendon or the long head of the biceps brachii. It alerts the user if they exceed an adjustable limit, predicting when they might injure themselves. In future iterations, the limit will be pre-calculated based on user characteristics and demographics. Furthermore, it has two real-time graphs: one that plots the raw values read via Bluetooth from the EMG sensor and a second rms-smoothed graph of the data. After a data set is collected, the user can click a button on the connected graphical user interface that displays the aforementioned graphs as well two new ones. The third applies a Fourier transform to the signal to convert it to a frequency domain, and the fourth plots the mean frequency calculated per second of data. The slope of a line of best fit of the fourth graph can be used to determine the user's current state of muscle fatigue, which can factor into an individual's risk of injury [1], [2]. A commercially-available and non-professional product for use during exercise such as the Myowearable will provide cost-friendly access to a tool that can greatly reduce the risk of recurring injury.

Introduction & Background

Doctors often recommend their injured patients avoid strenuous or physically demanding activity to prevent accidental injury. However, many patients are unsure of where that limit lies, until they have gone too far and damaged their muscle more. Muscle activation occurs when neuromuscular electrical impulses are received in the muscle from the brain via neural motor pathways. An electromyography (EMG) sensor is designed to detect and measure these electrical impulses reaching a muscle.

While electromyography has existed for decades, successfully diagnosing neuromuscular disorders and aiding in physical therapy, there are a few issues which limit its general use from the everyday person. First, sEMG electrode patches essentially wire the patient to a computer and therefore constrain movement to a fixed range of exercises. Furthermore, sports therapists and other medical personnel are typically in charge of



Figure 1: A version of the sleeve

placing electrodes and analyzing data, since such sEMG devices are not designed for public use. The Myowearable sleeve aims to fix this through an adjustable sleeve design, Bluetooth connection, which removes physical wires from the patient to the computer, and open-source data analysis. Through this, it is a wearable sEMG device that can aid a variety of individuals in their fitness and rehabilitation journey.

The sleeve uses an EMG sensor product called a MyoWare Sensor, distributed by SparkFun Learning Technologies, a company that makes educational circuitry, Arduino parts, and guides. Using that sensor, it measures two different types of muscle activity: determining if the muscle is using a lot of force, as indicated by a large magnitude of electrical activity, or looking at overall electrical activity over time if the muscle is rapidly getting fatigued.

Hardware

The MyoWare Muscle Sensor is a consumer and student-friendly surface EMG sensor for basic circuitry. Typical to most sEMG sensors, it is composed of three electrodes. This particular orientation is known as a bipolar configuration, in which a pair of electrodes detect a signal and compare it to a third electrode, known as the reference electrode. In order to obtain optimal results from using the MyoWare sensor, the rightmost electrode out of the pair should be placed on the widest part of the muscle, whereas the leftmost electrode should be along the muscle, closer to the tendon. The reference electrode should be placed on a bony part of the body, or on a muscle that is not being analyzed by the Myowearable⁵. Since the placement and security of the electrodes against the skin largely influences the data collected, the location of the MyoWare sensor was heavily considered when designing the Myowearable sleeve.

The sleeve works by processing the analog signal from the powered MyoWare sensor in an Arduino board. In the prototype models, the SparkFun "RedBoard" Arduino model was used. The continuous analog signal is stored as a variable in the RedBoard, which runs on an Arduino code included either in the software package or the physical board. The RedBoard then performs three functions with the value of the variable. Firstly, if the value of the measured neuroelectric stimulation surpasses a given value (550 by default), it outputs a signal to the buzzer which is the audio alert portion of the Myowearable. Secondly, when the box is connected via USB-B cable to a laptop, it sends the values to the Arduino Serial Monitor, which can directly be read and plotted there in real-time.



Figure 2: MyoWare Sensor on relaxed arm. Note the green light indicating power, but no red light indicating no muscle activity detected



Figure 3: MyoWare Sensor on flexed arm. Note the green and red lights indicating both power and detected muscle activity

Thirdly, and most importantly, it sends the value via an output wire to the bluetooth module where it is processed in Python for the corresponding application.

The Bluetooth module used is the HC-05 Wireless Bluetooth Module by DSD Tech. Due to a limitation in the HC-05 model, the Bluetooth module was only able to send values from 0 to 255. Because of this, all raw values from the sensor are divided by 8 before passing through the module. This number was determined experimentally as values read by the sensor range from 0 to 1000. The final value is coded to be outputted as a digital signal. Because of this, connecting the output signal of the sEMG sensor directly to the Bluetooth module did not work, and the Redboard is a necessary intermediary in this prototype.

The prerequisite to use the Bluetooth module is a configuration process carried out by a simple circuit and configuration code (written in Arduino language). The circuit has one pin of the module connected to 5V power, another connected to ground, a third connected to Pin 9 (serving as the RX [for data received via Bluetooth connection] pin) of the RedBoard/Arduino Uno Board, and a fourth pin connected to pin 10 (served as the TX [data to be transmitted via Bluetooth] pin).

From the serial monitor of the Arduino IDE to this circuit, developers are able to configure wireless modules with instructions known as AT commands to change features such as the connection password, binding location, and slave/master role⁸. These commands can usually be found online. The Arduino code used to send AT commands can be found in Appendix D.

Once the module was configured correctly, it was able to begin sending data. The first process that the software carries out is reading and interpreting the data received from the Bluetooth module, a task far more involved than it seems. By comparison to the values read on the serial monitor, it was found that the Bluetooth module transmits data in a mixture of hexadecimal codes, characters representing their ASCII values, and valueless errors.

The correlation between the transmitted data and the values it represented was realized through the comparison between the values printed in the Arduino IDE Serial Monitor, ASCII tables, and mathematically adjusted (integer division as well as a scaling factor were involved) values received. Once the different types of code were identified, there was difficulty in explaining to the Python syntax to ignore the characters appended to each value, a difficult task as data values were of different string lengths. For example, `b'x>'` meant 62, the ASCII value of `>` and `b'x04'` meant 4, the hexadecimal value. The final software package Python code includes the team's own decryption system (see lines 27-65 and 83-99 of the class file). The advantage of the encrypted values is that it will help with privacy issues; in

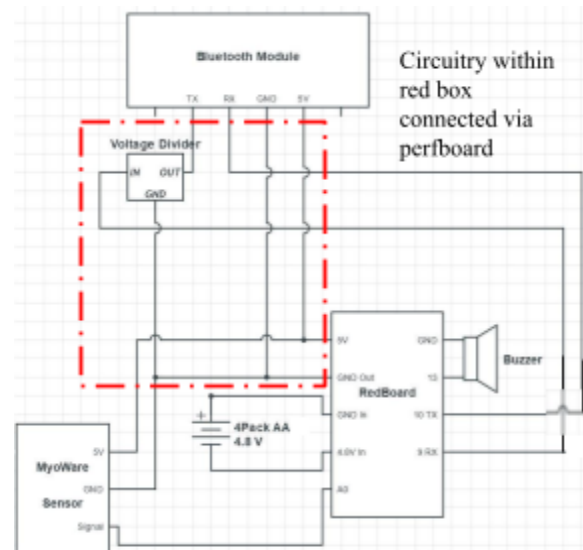


Figure 4: Final Circuit Schematic

the event of a malicious reception of the signal, the values will be totally unreadable.

The signal output from the RedBoard is passed through a voltage divider to decrease the signal from 5V to 3.3V, which the module is designed to take. The 5V goes into a split circuit (with 2K and 1K Ohm resistors) with one branch going into GND and the other into the bluetooth module. The remainder of the circuitry includes power and GND wires for the EMG sensor and Bluetooth module coming from the RedBoard, an unused wire from the Bluetooth module into the RedBoard for signal coming from a computer, and the power supply for the entire circuit.

Software

The software side of the product aims to receive the data sent from the Bluetooth module, plot it in real-time, carry out and output a variety of analyses, as well as save the data for future evaluation. The whole program is organized and displayed in a user-friendly graphical user interface (GUI), which was programmed using Python.

The received and deciphered data is graphed in real-time in two fields. The first graph shows the voltage (in millivolts) of the neuromuscular stimulation measured from each sample taken by the MyoWare sensor. The graph is limited to 200 points, corresponding to the most recent 5 seconds of data collected. The second graph is a smoothed version of the data in the first graph [3]. As for analysis, the Myowearable offers two methods of feedback for the user: a buzzer connected to the RedBoard, and a GUI button for graphing data. The buzzer relays real-time muscle activity information by playing when the user exceeds the signal threshold value, thereby notifying the user of the increased force in their movement. Along with the buzzer, a bar at the top of the GUI would turn red in color if the threshold is surpassed, or yellow if the incoming signal is approaching the threshold. Finally, if the user desires more detailed information regarding their exercise, they can click the data analysis button to open four graphs. The first two graphs are the sEMG data and RMS graphs from earlier, while the last two graphs are a discrete Fourier transform and plot of mean frequency over time.

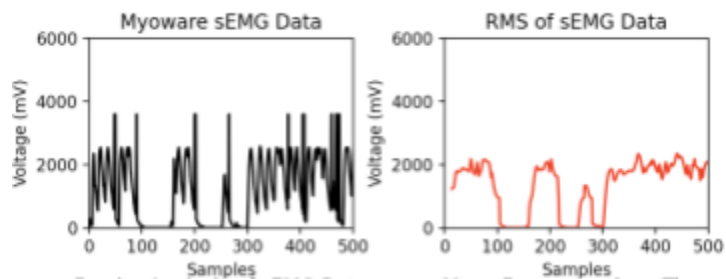


Figure 5: Raw Data Graph vs RMS Graph

Frequency-domain features, particularly the median and mean frequency, are often used in EMG analysis since a negative change in these parameters over time indicates the progression of fatigue [4]. In order to calculate these frequency parameters, the discrete Fourier transform is computed, as seen in the third graph. From performing a discrete Fourier transform on every chunk of data collected per second, the mean frequency per second was obtained and plotted over time (see Results & Discussion). By offering insight into how quickly one gets fatigued, the Myowearable can help its users understand the limits of their abilities in any exercise or rehabilitation routine.

Design



Figure 6: Sleeve oriented with bicep side upwards

The final Myowearable product consists of seven major parts: the sleeve, SparkFun RedBoard, MyoWare sensor, Bluetooth Module, a green case, soldered circuit board (perfboard), and Myowearable software. Additional features that were in the sleeve was a battery pack with 4 AA batteries, a piezo speaker, a voltage divider with resistors, and jumper cables connecting pins from the Redboard to the perfboard. The sleeve, RedBoard, MyoWare sensor, and Bluetooth module were purchased (from CVS, SparkFun Distributors, SparkFun Distributors, and DSD Tech, respectively). The case was 3D printed, the perf board had wires soldered in and then was connected to the redboard after the layout had been determined on a breadboard, and the Myowearable software was written to accommodate the Bluetooth module, graphing functions, and data analysis functions.

When coming up with a design for the sleeve, a constraint matrix and a decision matrix were created to help decide what features are valued more than others. These can be found in Appendix E. Based on the constraint matrix, it was concluded that making the sEMG device portable/Bluetooth operated while also having an efficient power source was the highest priority. Size, weight, adjustability, and minimal wiring was also important since it is essential to make the sleeve comfortable and wearable for people; the sleeve needs to work for many participants with different arm sizes, so adjustability is also quite essential.

The decision matrix was made to decide what part of the body would be best to measure muscle fatigue. The two best options were either measuring the semimembranosus hamstring muscle in the thigh or the the long head of the biceps brachii in the arm. Initially, a sleeve for the thigh was the main interest. However, it posed major challenges in terms of easy testing, and additional wiring. The wires can snap much easier when putting the sleeve on the thigh as opposed to the arm. Therefore, it was concluded that building the prototype for the arm would be best.

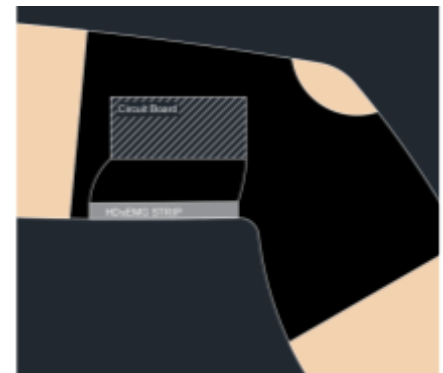


Figure 7: Original concept sketch of sleeve using the knee

During the creation process, the Myowearable underwent many design modifications to overcome problems with the code and sensor, better compile data, and for ease of user accessibility. First was the need for the SparkFun Redboard. Due to its size, it would make the design more bulky. However, developing a separate circuit board would require materials and resources that were not accessible to the authors in the First Year Learning Engineering Center. Due to the time restriction of completing a prototype, it would also require too much time for a company to deliver a printed circuit board.

With the addition of a redboard in the design, the power source needed to change as well. Since a connection to an outlet would cause dangling wires and limit movement, a battery had to be used. The RedBoard needs 5 volts to run the sleeve circuitry, so a 9 volt battery was considered.

Most Arduino boards have an exterior voltage input, with a suggested voltage range of 7-12V. This was the ideal option due to its size and easy replaceability. However, the boards use a linear regulator to reduce voltage, which is inefficient and battery draining. For example, if only 50mA is being drawn, this linear regulator consumes 0.2W, with the Arduino itself consuming 0.25W. 9 volt batteries are very low capacity as well; the capacity of a 9V Duracell battery is 580 mAh while the capacity of a single 1.5v Duracell AA battery is 2850 mAh. Therefore, despite the bulkier design, since a battery pack was accessible, 4 AA batteries are used as the power supply for this device [9].

In order to attach the circuitry to the sleeve, a box big enough to hold all the wires and circuitry was 3D printed. Although it made the product larger than intended, it also protected the circuitry from damage and movement. The addition of the extrusion for the switch and the buzzer makes the device more accessible to the user without having to completely open the entirety of the box. Additionally, the box is sewed to the sleeve via “feet” on the edges to ensure it stays intact through the various movements of the user. Interlocking parts were added to the lid and sides of the 3D box in order for the box to be securely closed. The only component outside of the box is the MyoWare sensor and its corresponding wires, which needs to be flush to the arm.

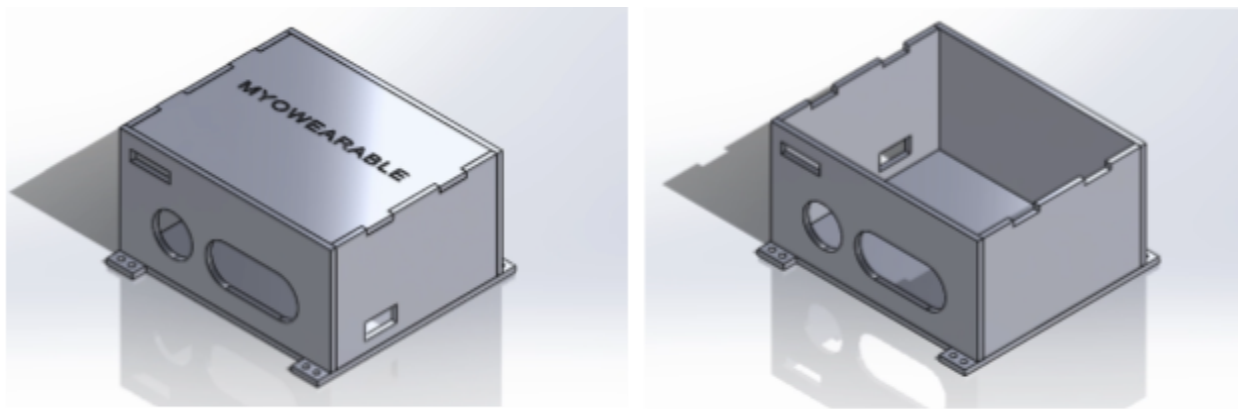


Figure 8: 3D Myowearable Box Design from SolidWorks This design includes holes for switch, USB connection, buzzer, and wires for Myoware Sensor.

Programming for the Myowearable also underwent major revisions. It was originally written in MATLAB due to easy access to the data of the MyoWare sensor and the vast data analysis capabilities. However, throughout the trials needed to connect the Bluetooth module on the circuit, it became evident that continuing with MATLAB would require a more difficult set up, and there was insufficient time to figure that out. Hence, the code was switched from MATLAB to Python for interfacing ease with Bluetooth, which utilizes sockets and the PyBluez2 module. This also allowed for the development of a GUI and a more complex programming component than initially designed.

Device Use & Data Collection Procedure

Using sEMG requires precise electrode placement and good skin contact for good measurements [10]. Hence, the Myowearable was designed to prioritize the position of the MyoWare sensor on

the muscle in a comfortable way. As the user slides the sleeve up their arm, they should ensure that the box containing circuitry is on their triceps, while the MyoWare sensor is on the top of their arm at the widest part of their bicep. The user should tighten or loosen the sleeve with the adjustable Velcro straps to ensure that the sensor does not shift position.

When collecting data for the Myowearable, the user would either rest their arm horizontally on a table or let their arm hang. Then, the participant would be asked to flex their arm close to a 90-degree angle, and then hold that position for a couple of seconds. Afterwards, they could return back to their resting position. After the user was at rest for a few more seconds, they were again told to flex their arm. This process would be repeated for at least a minute, as there would be sufficient data collected. The total data collected with the Myowearable contains 22 samples taken from students and faculty at Northeastern.

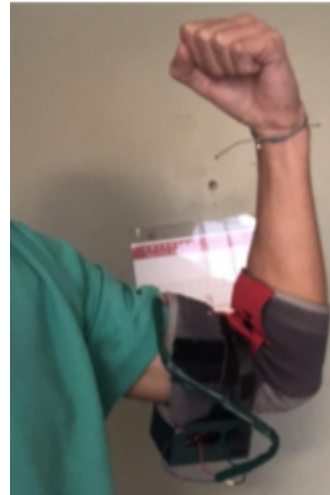


Figure 9: Sleeve measuring neuroelectrical signals telling bicep to contract

Results & Discussion

In order to draw conclusions on the user's susceptibility to injury, the Myowearable analyzes muscle activity through real-time biofeedback and fatigue analysis on demand. Muscle activity can be attributed to one of two types of movement, dynamic and isometric movement. In isometric movement, one can build strength by keeping their muscle in a fixed position, whereas in dynamic movement, the muscle is moving around and changing in length [11].

For dynamic movement analysis, every incoming signal is compared to a threshold value. If the signal value is higher than the threshold, the color bar on the GUI changes color from green to red while if the signal is close to the threshold the bar changes color from green to yellow. If the signal is well below the threshold, the bar remains green. The buzzer on the sleeve will notify the user of their current muscle activity by only buzzing when the most recent signal reaches or surpasses the threshold. This real-time warning works to prevent injury caused by maintaining that level of force, as the user would be reminded of where their limit of safe use is, and to reduce their effort for the remaining part of their exercise. It's important to note that the buzzing signal does not warn users of an imminent injury, but rather quantify and inform the user of their current force and strength.

As the user records data with the Myowearable, they will be presented with two graphs. The top graph plots the samples collected and their corresponding voltage as collected by the Myowearable. The sEMG data is received at a rate of 40 Hz, so the top graph displays the latest 200 values recorded, or 5 seconds worth of data. After the first 200 samples are recorded, a second plot graphs the filtered data against time. Because of this, the smoothed graph has a delay compared to real-time or the first graph. The MyoWare sensor currently returns a rectified EMG signal, which filters out negative voltages, so it was decided that one additional filter would be

used to process the EMG signal: a moving root mean square, which would operate on a window of 15 incoming data points to smooth out the graph. A moving RMS window summarizes the main features of the incoming data, gives insight into the average power of the signal, and ensures a positive output [12]. To compute the root mean square, the most recent 15 values are squared, and then their mean is calculated. The square root of the mean is then returned and plotted on the graph. This can be summarized with the following equation,

$$X_{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N |X_i|^2}.$$

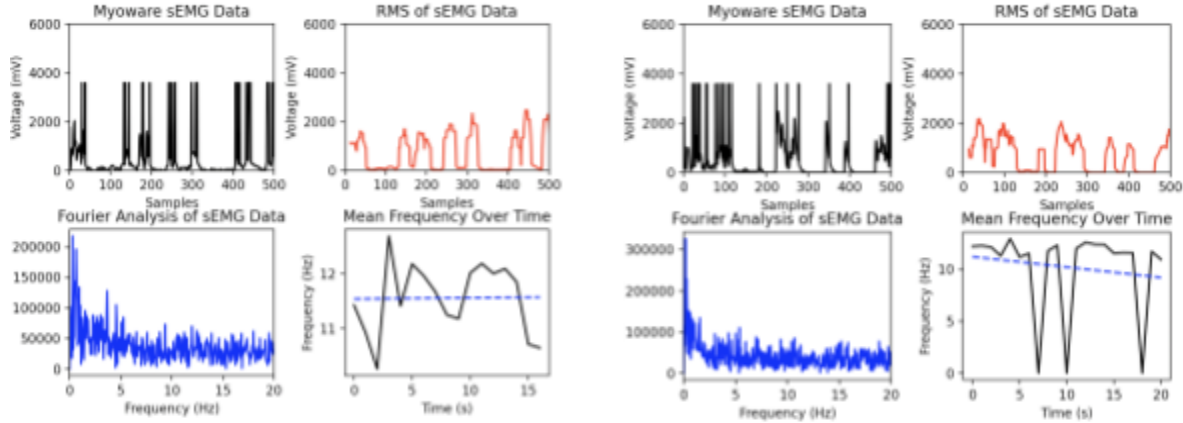


Figure 10: Sample data

The last two graphs outputted by the GUI are for the fatigue analysis conducted by the Myowearable. The third graph is a discrete Fourier transform that transforms the entire dataset from the time domain to a frequency domain in order to break the signal into its frequency components. The Fourier transform is used to calculate a power spectrum, which is needed to find mean frequency. The recorded data set is split into 1 second blocks, each containing 40 samples. For each block, a one-sided discrete Fourier transform (DFT) is computed. With the DFT per second, the power spectrum per second is calculated with the following formula,

$$P(f) = \frac{1}{N} \sum_{i=1}^N |x[n]e^{-j2\pi fn}|^2$$

in which it squares the absolute value of the signal's DFT [13]. The

mean frequency was calculated with the following equation, $MNF = \frac{\sum_{i=1}^N f_i P_i}{\sum_{j=1}^N P_j}$, which uses the

power spectrum created per second [4]. From the plot of mean frequency per second, a linear line of best fit was found and plotted on top. The value of the slope of the line is returned to the user, with a comment corresponding to the sign and magnitude of the slope. If the slope is negative and greater than -0.5 Hz/s, the user will be notified of a mild progression of fatigue. Furthermore, if the negative slope is steeper (less than or equal to -1 Hz/s), the user will be warned to consider the intensity of their past exercise as their peak ability due to the fast progression of fatigue demonstrated. On the other hand, if the slope is positive or near-constant, the user will be encouraged to keep exercising at the same rate, as it would demonstrate increased muscle activity with no decline in strength.

Conclusions & Implications

For individuals returning to regular exercise after an injury, it is crucial that they recognize the limits of their muscles to ensure complete recovery. With the Myowearable, a patient can benefit from audio and visual feedback, through the buzzer and GUI. While it won't predict injury, based on the user's movement the user can gain an understanding of how much force they are applying in an exercise and adjust accordingly. Similarly, if the user reviewed the graphs describing muscle activity and fatigue after attempting a few rehabilitation exercises, they could tailor a workout regimen, unique to their abilities, that could challenge them without the risk of muscle strain.

With more resources such as time and funding, numerous improvements could be implemented on the Myowearable to make it more user friendly, durable, and aesthetic. The first two improvements relate to general neatness and experience. Professional or mechanical soldering, as well as a printed circuit board, would make the circuitry neat, durable, and also shorten or eliminate much of the wiring. Similarly, a more comfortable sleeve could be developed by integrating a strapping system, which would replace the functional tape and Velcro straps used and ensure that the Myowearable could fit all arm sizes.

The lack of time furthermore inhibited the depth and accuracy of the data analysis conducted by the Myowearable. Signal processing is an incredibly nuanced field, so with more knowledge of the mathematics required to interpret data from the MyoWare sensor, the results offered by Myowearable could increase in reliability and usefulness. Increased experience in EMG signal processing would additionally allow for features like real-time fatigue analysis and gait analysis in the Myowearable, which would improve its versatility as an sEMG-based device.

Original designs included an IMU (inertial measurement units) sensor, but due to time constraints, size, and difficulty, it was omitted. In a more advanced version of the Myowearable sleeve, an IMU sensor could be added to collect additional data about the actual movement of the muscle. When the user does dynamic exercise, combining the IMU data with the EMG data would help the Myowearable analysis output a more informed conclusion regarding the strain of the user's muscle and the success of their muscle movement.

The largest improvement that could be made to the current design would be to obtain a small processor capable of translating analog signals from the MyoWare sensor into digital signals for the Bluetooth module to read. While the circuitry uses the internal capabilities of the RedBoard to do this, the RedBoard is bulky and requires a considerable amount of amperage to function, as it burned out and would not accept electricity from multiple battery sources. Without the RedBoard, the circuitry could run on a much smaller (~2mm by 15mm by 20mm) rechargeable Lithium Polymer (LiPo) battery, instead of the battery pack. As a result, the case would be much smaller and lighter, perfect for daily use and exercise.

Since the Myowearable intends on helping users understand their physical limitations, it is a device that can be used in fields like sports therapy. Similarly, its durability, reliability, immediate data analysis, and user-friendly interface also demonstrates its potential as a household fitness device. With future improvements, the Myowearable could also aid in research. For example, an experiment regarding the effect that aging has on the performance of

muscle function and duration of muscle fatigue could use the Myowearable to collect data on the muscle tension and fatigue in different age groups completing daily tasks, as opposed to exercising. Ultimately, the Myowearable is a promising medical device that can aid in various scenarios, ranging from personal, daily use to experimental and research use.

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Appendix A: Materials and Estimated Costs

Material	Estimated Cost	Material (cont.)	Estimated Cost (cont.)
Sparkfun Inventor's Kit & Northeastern Parts Kit (includes Jumper Wires, Redboard, Resistors, USB Plug, Breadboard, LEDs)	\$90.00 (<i>Given</i>)	4 AA Batteries	\$6.37 (<i>FYELIC</i>)
Dupont Wires (M/F, F/F, M/M Wires)	\$6.98 (<i>Have</i>)	Green Plastic (<i>For 3D Printing</i>)	\$3.00 (<i>FYELIC</i>)
Battery Pack	\$0.00 (<i>In Kit</i>)	Perfboard	\$1.75 (<i>FYELIC</i>)
Piezospeaker	\$0.00 (<i>In Kit</i>)	Soldering Wires	\$10.00 (<i>FYELIC</i>)
DSD TECH HC-05 2.0 Bluetooth Module	\$9.99	Soldering Machine	\$37.00 (<i>FYELIC</i>)
MyoWare Sensor	\$37.99 (<i>Given</i>)	Small Resistors	\$2.00 (<i>FYELIC</i>)
Nylon Sleeve	\$15.00	Plastic Thread, Safety Pins, Needle & Plastic Buttons	\$4.00 (<i>Have</i>)
Surface Electrode Patches	\$5.00 (<i>Have</i>)	Tape (Duct, Electric, Gorilla, Normal)	\$6.00 (<i>FYELIC</i>)
Velcro Straps & Belts	\$2.00 (<i>FYELIC</i>)	Northeastern Bottle Holder	\$0.00 (<i>Given</i>)

Software / Programs Used	Estimated Cost
Arduino IDE	(<i>Have</i>)
SolidWorks 2020	(<i>License Through Northeastern</i>)
MATLAB 2021	(<i>License Through Northeastern</i>)
Visual Studio Code: Python	(<i>License Free</i>)
Blender (Animation Software)	(<i>License Free</i>)

GitHub	(Account Free)
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Estimated Cost to Replicate (without any givens)	≈ \$237.08
Estimated Cost of What the Group Spent	

Some of the materials were given to the authors and not purchased.

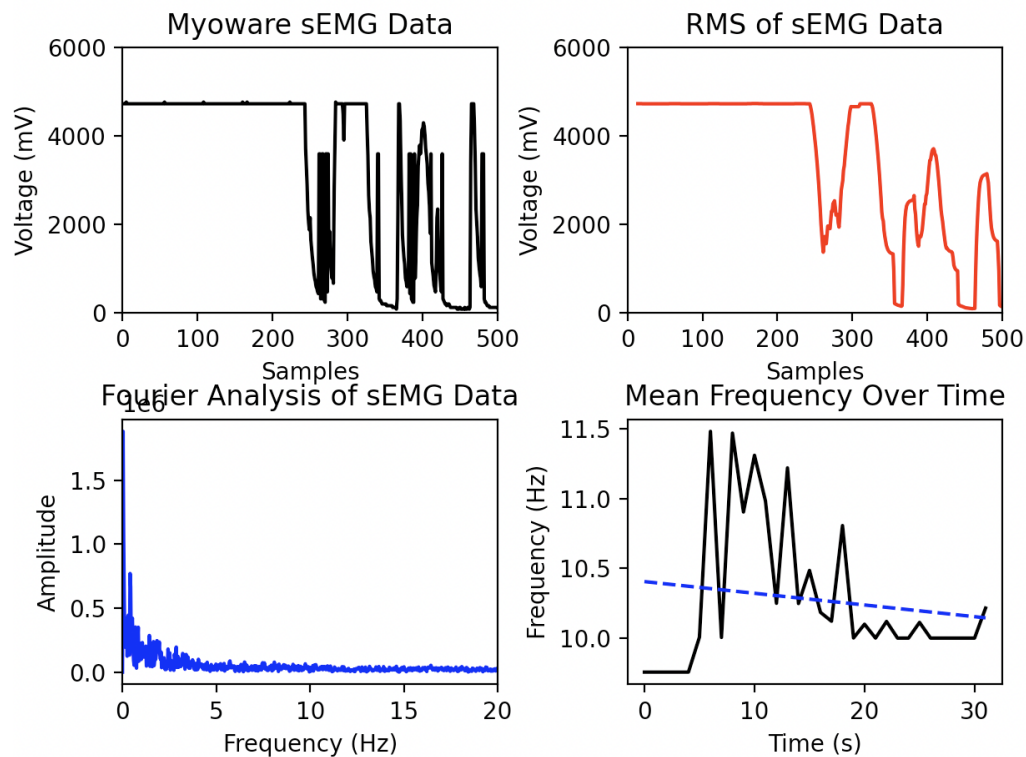
FYELIC stands for the First Year Engineering Learning & Innovation Center at Northeastern University.

Appendix B: 3 Samples of Data Collected

Sample 1

Rate of change in MNF: -0.008

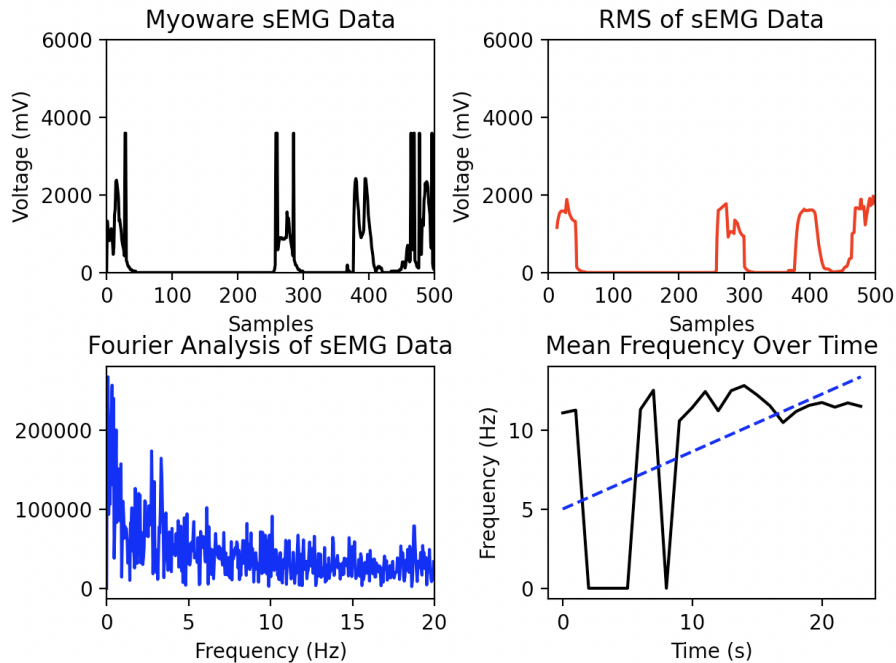
Slight progression of muscle fatigue. Keep going.



Sample 2

Rate of change in MNF: 0.363

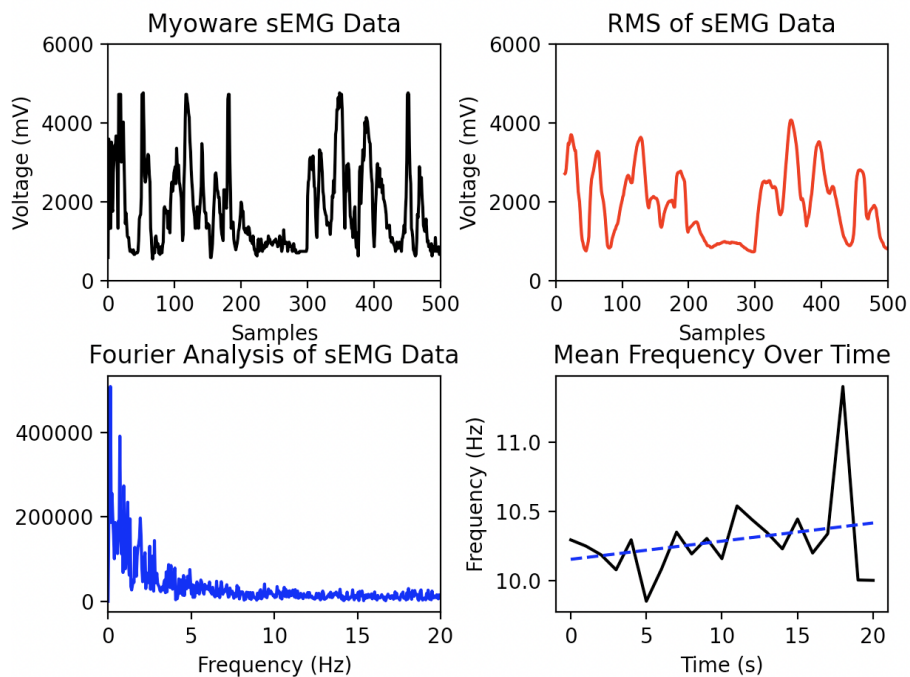
Increase in motor unit recruitment occurring. Keep up the good work.



Sample 3

Rate of change in MNF: 0.013

Negligible change in muscle fatigue. You can do it.



Appendix C: Circuit Component Images

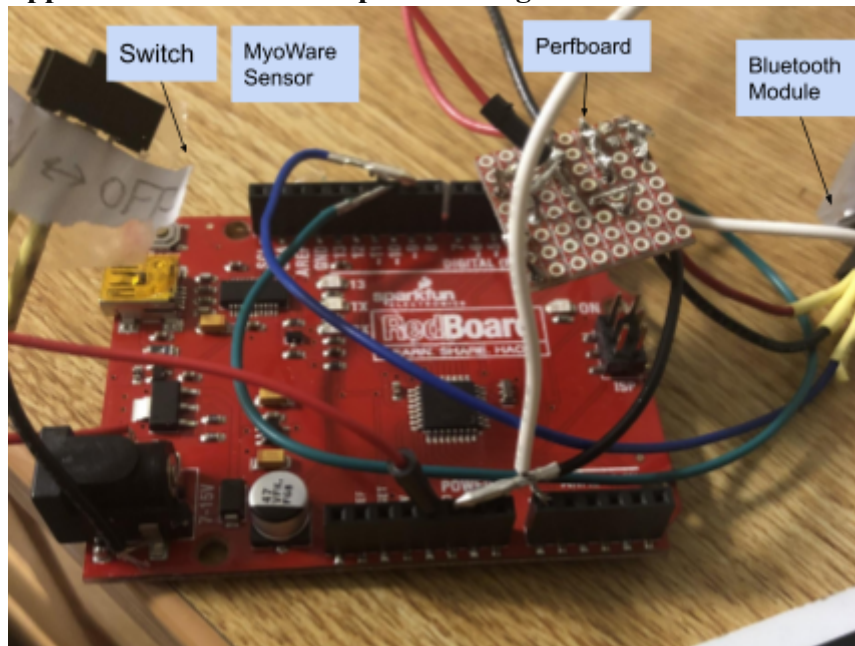


Image of 3D Printed Box with circuit components inside. Myoware Sensor is outside the box.



Image of circuitry components inside the 3D printed box. This includes the switch, wires connecting to various pins, piezospeaker, and the DSD TECH HC-05 Bluetooth module.



Image of 3D Printed Box with Battery Pack placed on top of circuit and redboard.



Final Sleeve Design: Includes Velcro Belt/Straps for Adjustability, Northeastern Logo, Myoware Sensor Wiring, and 3D Printed Box with the rest of the circuitry.



Appendix D: Source Code

All Source Code can be found in GitHub folder at:

<https://github.com/AnanyaT129/Cornerstone-Final-Project>

Appendix E: Constraint and Design Matrices

Table 1: Constraint Matrix

	Aesthetically Pleasing	Low Cost	Efficient Power Source	Portability (Bluetooth)	Minimal Wiring	Size/Weight	Comfort/Adjustability	Total
Aesthetically Pleasing	-	0	0	0	0	0	0	0
Low Cost	1	-	0	0	0	0	0	1
Efficient Power Source	1	1	-	0.5	0.5	1	0.5	4.5
Portability (Bluetooth)	1	1	0.5	-	1	0.5	1	5
Minimal Wiring	1	1	0.5	0	-	0.5	0.5	3.5
Size/Weight	1	1	0	0	0.5	-	0.5	3
Comfort/Adjustability	1	1	0.5	0	0.5	0.5	-	3.5

Table 2: Decision Matrix

Objectives	Aesthetically Pleasing	Low Cost	Efficient Power Source	Portability (Bluetooth)	Minimal Wiring	Size/Weight	Comfort/Adjustability	Total
Weights	20	30	90	100	70	60	70	-
Sleeve on Arm	4/80'	5/150'	6/540'	7/700'	4/280'	3/180'	4/280'	2,210
Sleeve on Thigh	2/40'	4/120'	6/540'	6/600'	2/140'	2/180'	2/140'	1,760